MUHAMMAD AHMED KHAN



Software Developer | Python, SQL & AI-Driven Data Analytics
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PROFESSIONAL SUMMARY

Master's student in Research in Computer and Systems Engineering at TU Ilmenau with over two years of hands-on experience as a software engineer in Python, SQL, ERP/hospital management systems and AI/NLP solutions. Skilled in optimizing SQL-based data workflows, data migration and validation, reports and dashboards, and chatbot and web/mobile applications, with a proactive, communicative working style built through cross-functional collaboration. Seeking a Working Student role where I can apply my Python, SQL and AI-tooling skills to support software performance analysis and innovative, sustainability-focused technology.

TECHNICAL SKILLS

Programming & Databases: Python, SQL (Query Optimization, Data Validation, Joins, Data Migration), JavaScript, SAP ABAP (Database Tables, Adobe Forms, F4 Help, Z_Reports)

AI Tools & Analytics: AI-driven data analytics, dashboards & reporting, NLP, Transformer Models (MT5), Cosine Similarity, KNN, Chatbot Development, Generative AI

Backend / Web: FastAPI, REST APIs, API Integration, React Native, HTML, CSS

Tools & Ways of Working: Git, GitHub, Selenium (Automation Testing), ERP & Hospital Management Systems, System Testing & Validation, Technical Documentation (SOPs), stakeholder communication

PROFESSIONAL EXPERIENCE

Junior Software Engineer | *The Kidney Centre (Post Graduate Training Institute), Karachi, Pakistan* Jul 2025 – Mar 2026

- Contributed to the development of in-house software systems with internal teams and external software partners.
- Worked on ERP and hospital management systems: implementation, testing, troubleshooting and continuous improvement.
- Supported software requirements for hospital departments, including pharmacy inventory, laboratory, emergency/clinical workflows and operational reporting.
- Used SQL extensively for database updates, data validation, troubleshooting and data consistency across modules; supported data migration, module testing and system validation during ERP implementation and software updates.
- Reviewed and optimized SQL queries to improve database performance and data retrieval (~25% faster retrieval on large datasets).
- Gathered and analyzed business requirements from departments and coordinated with developers and stakeholders to translate operational needs into software solutions.
- Documented system workflows and prepared SOPs and user guidelines, reducing onboarding time for new users.

Software Engineering Trainee – AI Engineer | *Dimensional Sys, Inc., Karachi, Pakistan* Feb 2025 – Apr 2025

- Developed AI-powered training chatbot solutions for different content formats, including PDF documents, videos and voice-based training material.
- Designed chatbot architectures and processing workflows connecting training content with AI-powered conversational interfaces.
- Worked with NLP and AI models so users can interact with training material through natural-language questions; applied NLP-based similarity algorithms to improve response relevance.
- Integrated APIs, AI services and application components; deployed and hosted chatbot applications and tested AI prototypes and proofs of concept, turning static and multimedia training content into interactive learning.

Junior Software Engineer | *Pinnacle Digitals, Karachi, Pakistan* Jun 2024 – Feb 2025

- Worked under senior supervision on ongoing projects: solving bugs, maintaining websites, and building and integrating new features using HTML, CSS and JavaScript.
- Worked independently on front-end and back-end tasks, including API integration and database management, and built a React Native Android app for the software house end to end.
- Debugged live projects, improved performance and cross-browser compatibility; automated website testing with Selenium and took part in code reviews.

SAP ABAP Developer Intern | *Systems Limited, Karachi, Pakistan* Aug 2023 – Nov 2023

- Created and managed database tables for efficient data storage, and merged multiple tables for improved data integration and retrieval.
- Designed Adobe Forms layouts to improve the presentation of documents within SAP.
- Implemented F4 Help and custom selection screens for tailored data retrieval and reporting; contributed to custom Z_Reports.

PROJECTS & RESEARCH

Urdu Paraphrasing Tool — AI/NLP Project

- Built a transformer-based (MT5) paraphrasing system for Urdu, a low-resource language, with a React Native app for real-world use.

Research Publication: "Training a Large Corpus Urdu Dataset for Paraphrasing on MT5"

- Published research on improving NLP performance for low-resource languages using large-scale datasets.

Exam Plagiarism Checker — AI System

- Built a plagiarism detection system using cosine similarity to identify academic dishonesty patterns.

EDUCATION

M.Sc. Research in Computer and Systems Engineering 2026 – Present
Technische Universität Ilmenau, Germany

B.Sc. Software Engineering 2020 – 2024
Sir Syed University of Science & Technology | CGPA: 3.49 / 4.00

CERTIFICATIONS

SQL Bootcamp: Zero to Advanced — Udemy (2025) • Software Quality & Automation — Sir Syed University

LANGUAGES

English (C1) • German (A1)